

University of Engineering and Technology, New Campus, Lahore
Department of Computer Science

Complex Computing Problem

Course Code & Title: CSC-451 Natural Language Processing

1. Course Learning Outcome:

CLO 3: Design and develop solution for syntax and semantic processing

Level of CLO: CLO-3 is mapped to the 6th level.

2. CCP Attributes:

- **A2 Depth of analysis required:** Has no obvious solution and requires conceptual thinking and innovative analysis to formulate suitable abstract models.
- **A3 Depth of knowledge required:** A solution requires the use of in-depth computing or domain knowledge and an analytical approach that is based on well-founded principles.
- **A8 Interdependence:** Is a high-level problem possibly including many component parts or subproblems.

3. Problem Statement:

Develop a machine translation system which converts the

1. Urdu sentence to English sentence
2. Roman Urdu to Urdu

Problem Description:

This project explores the application of machine translation techniques for Roman Urdu to Urdu and Urdu to English. Students will develop and analyse dataset to develop machine translation models and evaluate its performance on testing data.

- Your task is to:
 - Create and Analyse Dataset:
 - Urdu to English: Each group will download 1 lac sentences having parallel English translation. English translation has been generated using google translation tool. Each member of the group will verify the sentences for correct translation.
 - Roman Urdu to Urdu: Each group will run the provided utility and generate parallel Roman Urdu sentences of the verified 1 lac sentences.
 - Model Selection and implementation:
 - Choose a suitable 1 deep learning model for machine translation.
 - Train or Fine-tune the model on developed dataset.

- Experiment with hyperparameter tuning to optimize model performance.
- Implement Zero-shot and few-shot prompting for the state of the art TWO large language models
- Evaluate and Analyze:
 - Evaluate the model's performance using automated metrics like BLUE and/or ROUGE scores.
 - Extract and present error analysis for each model separately (TWO LLMS)
 - Zero Shot
 - Few Shot
 - Compare results with Fine-tuned model

4. Instructions:

- This is a group project having 3-4 students .
- Your project will be graded according to the provided rubrics.
- Deadline to submit: 1st January, 2025

5. Deliverables:

- A detailed report outlining approach, including data pre-processing steps, model selection, hyperparameter tuning strategies, and evaluation results.
- The trained/fine-tuned model and code used for training and evaluation.

6. Alignment with Attributes of Complex Computing Problems:

Attribute	Alignment / Justification
Depth of analysis required	• Data development for Roman Urdu-to-Urdu and Urdu-to-English translation would be from scratch. Students will analyse that rule based model cannot be applied, huge dataset is required to have coverage • Requires conceptual thinking and innovative analysis to search and select the respective deep learning model for the machine translation system.
Depth of knowledge required	• Use of in-depth computing ○ Data preprocessing: uppcase, lower case normalization, stop words removal, stemmer, Unicode normalization etc ○ Model selection: model training and testing

	○ Domain knowledge for machine translation for RomanUrdu-to-Urdu and Urdu-to-English challenges and respective solution. • An analytical approach for encoding translation data(embedding preferred, selecting suitable models, and normalizing data etc.
Interdependence	• Data preprocessing • Models (2-3 models) • Testing and evaluation • Automatic analysis module

Complex Computing Problem Assessment Rubrics

Criteria & CCP Attributes	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Poor (1)
Data Development & Cleaning (A2, A3)	Dataset is complete cleaned and manually reviewed	Data completeness and cleaning is 80% done	Data completeness and cleaning is 60% done	Data completeness and cleaning is 40% done	Very small subset of dataset is developed No cleaning process is applied.
Data Preprocessing (A2, A3)	Normalization, stop words removal, stemmer and Embedding are implemented and have complete understanding	Normalization, stop words removal, stemmer and Embedding (Embedding must) 80% Completed and have understanding	Normalization, stop words removal, stemmer and Embedding (Embedding must) 60% Completed and have understanding	Normalization, stop words removal, stemmer and Embedding (Embedding must) 40% Completed and have understanding	Minimal preprocessing steps implemented
Model Selection & Implementation (A3, A8)	1 deep learning models training, and Zero and Few Shot Prompting of TWO Models and testing done.	1 deep learning models training, and Zero and Few Shot Prompting of ONE Models and testing done.	1 deep learning models Training and testing done.	Try to implement model	No model trained and tested 0 mark

Criteria & CCP Attributes	Excellent (5)	Good (4)	Satisfactory (3)	Needs Improvement (2)	Poor (1)
Model Evaluation & Metrics (A3)	BLUE score and ROUGE scores are implemented and comparisons of 3 models is also presented in the form of table	Either 1 evaluation metric missing OR two models comparisons available	Both Evaluation metrics are implemented but results of one model are not presented	Evaluation metrics are implemented but results are not presented	No evaluation metric is implemented
Analysis and Discussion (A2)	Proper analysis with examples of good translations Average translations Worst translations	Examples of any category is missing (best, average and bad)	Average analysis report: Minimal examples may be any category is missing (best, average and bad)	Weak analysis	NO analysis